

Topic I: Balanced Trees – Conceptual AVL & Rotations

Balance Factors, Single/Double Rotations, LL/RR/LR/RL Invariants, BST Properties

Section 1: Previous Year Questions (PYQs 1 to 3)

Q1. Permissible AVL Balance Factors

GATE CS 2004 / DA Benchmark | MCQ

In an AVL tree, let the balance factor of any node u be defined as $BF(u) = \text{height}(\text{left_subtree}) - \text{height}(\text{right_subtree})$. Which of the following sets represents the strictly permissible values for $BF(u)$?

- (A) $\{-1, 0, 1\}$ (B) $\{0, 1, 2\}$
(C) $\{-2, -1, 0, 1, 2\}$ (D) $\{0, 1\}$

Answer: (A) Step-by-step Solution: By the formal definition of an Adelson-Velsky and Landis (AVL) tree, every node must satisfy $|\text{height}(\text{left}) - \text{height}(\text{right})| \leq 1$. Thus, the only legal balance factors are $-1, 0$, and $+1$. If any node reaches ± 2 after an insertion or deletion, a rotation is triggered immediately.

Q2. Single vs Double AVL Rotation Selection

GATE CS 2014 | MCQ

An AVL tree experiences a balance factor violation of $+2$ at node A after inserting a key into the right subtree of A 's left child. To restore the AVL property, which rotation must be performed?

- (A) Single Left (LL) rotation (B) Single Right (RR) rotation
(C) Double Left-Right (LR) rotation (D) Double Right-Left (RL) rotation

Answer: (C) Step-by-step Solution: Node A is left-heavy ($+2$). The newly inserted node is in the *right* subtree of A 's *left* child (LR case). An LR imbalance cannot be fixed by a single rotation; it requires a Left rotation on the left child followed by a Right rotation on node A .

Q3. Rotation Invariance on Inorder Sequence

GATE CS 2008 | MCQ

When any tree rotation (LL, RR, LR, or RL) is executed on an AVL binary search tree, what happens to the inorder traversal sequence of the keys?

- (A) It is reversed. (B) It remains completely unchanged.
(C) The root moves to the first position. (D) It becomes dependent on the height.

Answer: (B) Step-by-step Solution: Rotations are local restructuring operations designed specifically to adjust subtree heights while preserving the fundamental binary search tree ordering invariant. Inorder traversal of any valid BST lists elements in ascending sorted order; hence, the inorder sequence remains strictly invariant under all rotations.

Topic II: Tree Height & Balance Complexity (PYQs 4 to 6)

Q4. Minimum Nodes in an AVL Tree of Height h

GATE CS 2002 / 2016 | MCQ

Let $N(h)$ denote the minimum number of nodes in an AVL tree of height h , with base conditions $N(0) = 1$ and $N(1) = 2$. The recurrence relation governing $N(h)$ for $h \geq 2$ is:

- (A) $N(h) = N(h-1) + N(h-2) + 1$ (B) $N(h) = 2N(h-1) + 1$
(C) $N(h) = N(h-1) + N(h-2)$ (D) $N(h) = 2N(h-2) + 1$

Answer: (A) Step-by-step Solution: To minimize nodes while achieving height h , one subtree must have height $h-1$ and the other must have minimum permissible height $h-2$. Including the root node gives the recurrence: $N(h) = N(h-1) + N(h-2) + 1$. This Fibonacci-like relation proves that AVL height is strictly bounded by $O(\log n)$.

Q5. Minimum Nodes for AVL Tree of Height 4

GATE CS 2013 | NAT

Taking the height of a single-node AVL tree as 0, what is the minimum number of nodes required to construct an AVL tree of height 4?

Answer: 12 Step-by-step Solution: Using $N(h) = N(h-1) + N(h-2) + 1$:

- $h = 0$: $N(0) = 1$
- $h = 1$: $N(1) = 2$
- $h = 2$: $N(2) = 2 + 1 + 1 = 4$
- $h = 3$: $N(3) = 4 + 2 + 1 = 7$
- $h = 4$: $N(4) = 7 + 4 + 1 = 12$.

Q6. Maximum Height of an AVL Tree with n Nodes

GATE CS 2007 | MCQ

What is the upper bound on the height h of an AVL tree containing n elements as $n \rightarrow \infty$?

- (A) $\approx 1.44 \log_2 n$ (B) $\approx 2.00 \log_2 n$
(C) $\approx 0.50 \log_2 n$ (D) $\approx \log_{10} n$

Answer: (A) Step-by-step Solution: Because $N(h) \approx (1/\sqrt{5}) \cdot \Phi^{h+2} - 1$ where the golden ratio $\Phi \approx 1.618$, taking logarithms gives $h \leq (1/\log_2 \Phi) \cdot \log_2 n \approx 1.4404 \log_2 n$. Hence, search/insert/delete in AVL trees is guaranteed to be $O(\log n)$ in the absolute worst case.

Q7. Logical Entailment and Unsatisfiability Principle

GATE DA 2024 / CS Core | MCQ

In propositional logic, a knowledge base KB logically entails a sentence α (denoted $KB \models \alpha$) if and only if which of the following compound sentences is **UNSATISFIABLE**?

- (A) $KB \wedge \neg\alpha$ (B) $KB \wedge \alpha$
(C) $KB \vee \neg\alpha$ (D) $\neg KB \wedge \alpha$

Answer: (A) Step-by-step Solution: By the refutation completeness theorem (proof by contradiction): $KB \models \alpha$ holds iff in every model where KB is true, α is also true. Thus, there can exist no model where KB is true and α is false. Consequently, $(KB \wedge \neg\alpha)$ must have no satisfying truth assignment (must be unsatisfiable).

Q8. Tautology (Validity) Identification

GATE CS 2012 | MCQ

Which of the following propositional formulas is a **valid formula (tautology)**?

- (A) $((P \rightarrow Q) \wedge P) \rightarrow Q$ (B) $((P \rightarrow Q) \wedge Q) \rightarrow P$
(C) $(P \vee Q) \rightarrow (P \wedge Q)$ (D) $(P \rightarrow Q) \rightarrow (\neg P \rightarrow \neg Q)$

Answer: (A) Step-by-step Solution: (A) is the classic inference rule *Modus Ponens*. If P is true and $P \rightarrow Q$ is true, Q must be true. Assume (A) is False: then LHS $((P \rightarrow Q) \wedge P)$ is True and RHS (Q) is False. If Q is False and P is True, $P \rightarrow Q$ is False, which contradicts LHS being True. No falsifying assignment exists; hence it is valid. (B) is the fallacy of affirming the consequent.

Q9. Satisfiable but Not Valid Formulas

GATE CS 2015 | MSQ

A propositional formula is defined as *satisfiable* if it evaluates to True under at least one interpretation, and *valid* if it evaluates to True under all interpretations. Which of the following formulas is/are **satisfiable but NOT valid**? [Select all that apply]

- (A) $P \vee \neg P$ (B) $P \rightarrow \neg P$
(C) $P \wedge \neg P$ (D) $P \leftrightarrow Q$

Answer: (B) and (D) Step-by-step Solution: (A) is a tautology (valid). (C) is a contradiction (unsatisfiable). (B) evaluates to True when $P = \text{False}$ ($\text{False} \rightarrow \text{True}$ is True), and False when $P = \text{True}$; hence it is satisfiable but not valid. (D) evaluates to True when $P = Q$ and False when $P \neq Q$; hence it is also satisfiable but not valid.

Q10. Evaluating Three Formulas for Validity

GATE CS 2006 / DA Standard | MCQ

Consider the following three propositional formulas:

$$F_1: (P \wedge (P \rightarrow Q)) \rightarrow Q$$

$$F_2: (P \vee Q) \wedge (\neg P \vee \neg Q)$$

$$F_3: (P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$$

Which of the following assertions is **CORRECT**?

- (A) F_1 and F_3 are valid; F_2 is satisfiable but not valid. (B) F_1 , F_2 , and F_3 are all valid.
(C) F_2 is unsatisfiable; F_1 and F_3 are valid. (D) None of the three formulas is valid.

Answer: (A) **Step-by-step Solution:** F_1 is Modus Ponens (valid/tautology). F_3 expresses contraposition equivalence, which is logically always True (valid/tautology). F_2 is XOR: it is True when $P = \text{True}$, $Q = \text{False}$ (satisfiable), but False when $P = Q = \text{True}$ (not valid). Hence, F_1 and F_3 are valid, while F_2 is satisfiable but not valid.

Q11. Minimum Nodes for AVL Tree of Height 5

New Practice | NAT

Let the height of a single-node tree be 0. What is the minimum number of nodes in an AVL tree of height 5?

Answer: 20 **Step-by-step Solution:** Using $N(h) = N(h-1) + N(h-2) + 1$ with $N(0) = 1$, $N(1) = 2$:

$$N(2) = 2 + 1 + 1 = 4$$

$$N(3) = 4 + 2 + 1 = 7$$

$$N(4) = 7 + 4 + 1 = 12$$

$$N(5) = N(4) + N(3) + 1 = 12 + 7 + 1 = 20.$$

Q12. Insertion Rotation Sequence in an AVL Tree

New Practice | MCQ

Starting with an initially empty AVL tree, the keys **20, 10, 15** are inserted in that exact sequence. Which rotation is performed to restore balance, and which key becomes the root?

- (A) Single Right (RR) rotation; key 10 becomes root (B) Double Left-Right (LR) rotation; key 15 becomes root
(C) Single Left (LL) rotation; key 20 remains root (D) Double Right-Left (RL) rotation; key 10 becomes root

Answer: (B) **Step-by-step Solution:** (1) Insert 20 (root). (2) Insert 10 (left child of 20). (3) Insert 15: goes into right subtree of 10. Root 20 has balance factor = 2 (left subtree height 1, right subtree height -1 $\Rightarrow$ BF = +2). The insertion path is Left then Right (LR imbalance). A double LR rotation is executed: left-rotate on 10 promotes 15, then right-rotate on 20 places 15 as root, with 10 as left child and 20 as right child.

Practice Set: AVL Rotation Mechanics & Height Bounds (Practice 13 to 15)

Q13. AVL Tree Insertion Cost and Rotations Required

New Practice | MCQ

When a new key is inserted into an arbitrary AVL tree with n nodes, what is the maximum number of tree rotations (where a double rotation counts as 1 compound operation) required to restore the AVL balance property?

- (A) At most 1
(B) At most $O(\log n)$
(C) At most $n / 2$
(D) Exactly 2

Answer: (A) Step-by-step Solution: During an *insertion* into an AVL tree, rebalancing the lowest unbalanced ancestor with a single (LL or RR) or double (LR or RL) rotation restores the subtree height to what it was before insertion. Hence, no further ancestors are unbalanced, and at most 1 rotation operation suffices. (In contrast, *deletion* may require up to $O(\log n)$ rotations).

Q14. Balance Factor Distribution in Worst-Case Sparse AVL Trees

New Practice | MSQ

In a minimally populated AVL tree (Fibonacci tree) of height $h \geq 2$ containing the minimum possible number of nodes, which of the following statements is/are **TRUE**? [Select all that apply]

- (A) The root node has balance factor ± 1 .
(B) Every internal node has balance factor ± 1 .
(C) The number of leaves is equal to Fibonacci number F_{h+1} .
(D) The tree contains at least one node with balance factor 0.

Answer: (A), (B), and (C) Step-by-step Solution: To achieve minimum nodes for height h , every single internal node must have subtrees differing in height by exactly 1 (BF = +1 or -1). If any internal node had BF = 0, both subtrees would have the same height, which would require strictly more nodes. Leaf count follows the Fibonacci sequence F_{h+1} . Thus (A), (B), and (C) are true.

Q15. Maximum Nodes in AVL Tree of Given Height

New Practice | NAT

What is the **maximum** number of nodes that can be accommodated in an AVL tree of height 3 (height of a single root node is 0)?

Answer: 15 Step-by-step Solution: The maximum number of nodes in any binary tree (and hence in an AVL tree) of height h occurs when every level from 0 to h is completely filled (a perfect binary tree). For height $h = 3$, max nodes = $2^{h+1} - 1 = 2^4 - 1 = 16 - 1 = 15$. Every node in a perfect binary tree has balance factor 0, which trivially satisfies the AVL property.

Practice Set: Propositional Logic – Equivalence & Entailment (Practice 16 to 18)

Q16. Checking Entailment via Truth Assignment

[New Practice | MCQ](#)

Given the Knowledge Base consisting of two premises: $KB = \{P \vee Q, \neg Q \vee R\}$. Does KB logically entail $(\neg P \rightarrow R)$?

- (A) Yes, because in every model where KB is true, $(\neg P \rightarrow R)$ is also true. (B) No, because $P = \text{False}, Q = \text{True}, R = \text{False}$ is a counter-model.
(C) No, because resolution yields an empty clause. (D) It cannot be determined without knowing the truth value of Q .

Answer: (A) Step-by-step Solution: $\neg P \rightarrow R$ is logically equivalent to $\neg(\neg P) \vee R = (P \vee R)$. Applying propositional resolution on the two clauses in KB : resolve $(P \vee Q)$ with $(\neg Q \vee R)$ on complementary literal Q yields the resolvent $(P \vee R)$. Since the resolvent is derived directly by sound inference, $KB \models (P \vee R)$.

Q17. Satisfying Models Count for Propositional Formula

[New Practice | NAT](#)

How many distinct truth assignments (models) over the 3 boolean variables P, Q, R satisfy the formula: $(P \wedge Q) \rightarrow R$?

Answer: 7 Step-by-step Solution: A conditional statement $A \rightarrow B$ is False if and only if A is True and B is False. Here, LHS $(P \wedge Q)$ is True only when $P = \text{True}$ and $Q = \text{True}$. RHS R is False when $R = \text{False}$. Thus, the only truth assignment making the implication False is $(P = \text{True}, Q = \text{True}, R = \text{False})$ – exactly 1 assignment. Total possible assignments over 3 variables is $2^3 = 8$. Satisfying models = $8 - 1 = 7$.

Q18. Validity Properties under Logical Implication

[New Practice | MSQ](#)

Let α and β be arbitrary propositional formulas. Which of the following assertions is/are **ALWAYS TRUE**? [Select all that apply]

- (A) If α is valid and $(\alpha \rightarrow \beta)$ is valid, then β is valid. (B) If α is satisfiable, then $\neg\alpha$ must be unsatisfiable.
(C) α is valid if and only if $\neg\alpha$ is unsatisfiable. (D) If $(\alpha \wedge \beta)$ is satisfiable, then α is satisfiable and β is satisfiable.

Answer: (A), (C), and (D) Step-by-step Solution: (A) is the soundness of Modus Ponens: if α is true in all models and $\alpha \rightarrow \beta$ is true in all models, β must be true in all models. (C) is the duality principle of validity and unsatisfiability. (D) If an assignment makes both true simultaneously, it obviously satisfies each individually. (B) is false: for contingent α (e.g., $\alpha = P$), both P and $\neg P$ are satisfiable.

Topic IV Practice: Formulas Validity/Satisfiability Evaluations (Practice 19 to 21)

Q19. Three-Formula Evaluation Suite: Set A

[New Practice | MCQ](#)

Classify each of the following three formulas as [Valid / Satisfiable but not valid / Unsatisfiable]:

- $\varphi_1: ((P \rightarrow Q) \wedge (Q \rightarrow R)) \rightarrow (P \rightarrow R)$
- $\varphi_2: (P \vee Q) \wedge (\neg P \vee R) \wedge (\neg Q \vee R) \wedge \neg R$
- $\varphi_3: (P \wedge \neg Q) \rightarrow (\neg P \vee Q)$

(A) φ_1 is Valid; φ_2 is Unsatisfiable; φ_3 is Satisfiable but not valid.

(B) φ_1 is Valid; φ_2 is Satisfiable but not valid; φ_3 is Unsatisfiable.

(C) All three formulas are Satisfiable but not valid.

(D) φ_1 and φ_3 are Valid; φ_2 is Unsatisfiable.

Answer: (A) Step-by-step Solution: • φ_1 is Hypothetical Syllogism (transitivity of implication), which is a classical tautology (Valid).

• φ_2 : Since $\neg R$ is True, $R = \text{False}$. Then $(\neg P \vee R)$ forces $P = \text{False}$, and $(\neg Q \vee R)$ forces $Q = \text{False}$. But $(P \vee Q)$ requires at least one to be True. Contradiction! No truth assignment satisfies φ_2 (Unsatisfiable).

• φ_3 : RHS is $\neg(P \wedge \neg Q)$. Thus φ_3 has the form $X \rightarrow \neg X$. This is True when X is False (satisfiable), but False when X is True (not valid).

Q20. Three-Formula Evaluation Suite: Set B

[New Practice | MSQ](#)

Consider the following three propositional formulas:

- $\psi_1: (P \leftrightarrow Q) \leftrightarrow (\neg P \leftrightarrow \neg Q)$
- $\psi_2: (P \rightarrow (Q \rightarrow R)) \leftrightarrow ((P \wedge Q) \rightarrow R)$
- $\psi_3: (P \rightarrow Q) \vee (Q \rightarrow P)$

Which of these formulas is/are **VALID (tautologies)**? [Select all that apply]

(A) ψ_1

(B) ψ_2

(C) ψ_3

(D) None of them

Answer: (A), (B), and (C) Step-by-step Solution: • $\psi_1: (\neg P \leftrightarrow \neg Q)$ is identical to $(P \leftrightarrow Q)$. An expression biconditional with itself is always True (Valid).

• ψ_2 : This is the standard *Exportation law*: $P \rightarrow (Q \rightarrow R) \equiv \neg P \vee \neg Q \vee R \equiv (P \wedge Q) \rightarrow R$. Being an equivalence, the biconditional is always True (Valid).

• $\psi_3: (\neg P \vee Q) \vee (\neg Q \vee P) = (P \vee \neg P) \vee (Q \vee \neg Q) = \text{True} \vee \text{True} = \text{True}$ (Valid). All three are tautologies!

Q21. Determining Validity of Distributive Implication

[New Practice | MCQ](#)

Consider the formula: $(P \rightarrow (Q \vee R)) \rightarrow ((P \rightarrow Q) \vee (P \rightarrow R))$. Which of the following correctly describes this formula?

(A) Valid (Tautology)

(B) Satisfiable but not valid

(C) Unsatisfiable (Contradiction)

(D) Contingent only when P is false

Answer: (A) Step-by-step Solution: Rewrite RHS: $(\neg P \vee Q) \vee (\neg P \vee R) = \neg P \vee Q \vee R$. Rewrite LHS: $P \rightarrow (Q \vee R) = \neg P \vee (Q \vee R) = \neg P \vee Q \vee R$. The antecedent and consequent are syntactically and semantically identical. Any formula of the form $A \rightarrow A$ is unconditionally True for all interpretations, hence it is **Valid**.

Q22. Truth Table Models Count for Unsatisfiability

[New Practice | NAT](#)

A propositional logic sentence over 4 atomic variables A, B, C, D is expressed in Conjunctive Normal Form (CNF) as a conjunction of 16 distinct maxterms (each containing all 4 variables). How many satisfying models does this formula possess?

Answer: 0 **Step-by-step Solution:** With 4 boolean variables, there are $2^4 = 16$ possible truth assignments (interpretations). Each distinct maxterm is False for exactly 1 unique interpretation and True for the other 15. A conjunction of all 16 unique maxterms is False for every single interpretation. Therefore, the formula has exactly 0 satisfying models (it is a contradiction).

Q23. AVL Deletion Rotation Count

[New Practice | MCQ](#)

In the worst case, deleting a single node from an AVL tree containing n nodes may require how many rotation operations to restore balance all the way to the root?

- (A) At most 1 (B) At most 2
(C) $O(\log n)$ (D) $\Theta(n)$

Answer: (C) **Step-by-step Solution:** Unlike insertion (where a single rotation operation restores the original subtree height and terminates the rebalancing propagation), deletion may decrease the height of the rebalanced subtree by 1. This height decrease can propagate upwards and create imbalances at successive parent nodes, potentially requiring rotations at every level up to the root, which is $O(\log n)$ rotations.

Q24. Entailment with Monotonicity in Propositional Logic

[New Practice | MSQ](#)

Propositional logic is known to be **monotonic**. Which of the following mathematical statements correctly expresses the property of monotonicity? [Select all that apply]

- (A) If $KB \models \alpha$, then $(KB \wedge \beta) \models \alpha$ for any arbitrary sentence β . (B) If $KB \models \alpha$, adding new axioms cannot invalidate the entailment of α .
(C) If $KB \models \alpha$, then $KB \models \neg\alpha$ must be satisfiable. (D) If $KB \models (\alpha \vee \beta)$, then $KB \models \alpha$ or $KB \models \beta$.

Answer: (A) and (B) **Step-by-step Solution:** Monotonicity means the set of entailed sentences can only grow, never shrink, as information is added. Formally, if $KB \models \alpha$, then $(KB \wedge \beta) \models \alpha$. Adding new premises β cannot retract previously proven conclusions (making A and B correct). (D) is false: for instance, if $KB = \{P \vee Q\}$, KB entails $P \vee Q$, but entails neither P nor Q alone.